NEXT ENGINEER · _SHEERAT_

Design Thinking

BOE077

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 7 | **Credits:** 3 | **L-T-P:** 3-0-0

Scheme: AKTU B.Tech 4th year, Open Elective-II, 2025-26 onwards

UNIT 1 Introduction to Design Thinking

UNIT 2 Human Needs, Values and Empathy

UNIT 3 Problem Definition and Ideation

UNIT 4 Critical Thinking

UNIT 5 Arguments, Logic and Reasoning

Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: frameworks, tools and comparison tables.
3. **Examples and comparisons**: worked design-thinking cases (empathy map, journey map, POV and How Might We, 5 Whys, Six Thinking Hats, argument diagrams, fallacy spotting, truth tables).
4. **Key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (what design thinking is, traditional problem solving vs design thinking, history, wicked problems, innovation and creativity, design mindset, elements and principles of design) and Unit 2 (human needs and values, culture, stakeholders, empathy tools).
- **Afternoon (3 hours):** Unit 3 (problem statements, personas, POV, research, ideation tools, double diamond, 5 Whys, How Might We, mind mapping, Six Thinking Hats) and Unit 4 (critical thinking, traits, skills, pillars, argument vs rhetoric, cognitive biases).
- **Evening (2 hours):** Unit 5 (arguments, premises, conclusions, validity, soundness, deductive and inductive reasoning, argument diagrams, fallacies, propositional logic, probability and judgement). Then the "Last-hour checklist".
- **In the exam:** answer with a real example every time (a canteen queue, a campus app, a water bottle); draw the five-stage model and the double diamond; for logic questions draw the truth table.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Introduction to Design Thinking

✍️ 2-mark definitions

- **Design thinking:** a human-centred, iterative approach to solving problems that starts from understanding users' needs, reframes the problem, generates many ideas, and tests prototypes with users.
- **Human-centred design:** design that begins with the people you are designing for and ends with solutions tailored to their needs.
- **Wicked problem (Rittel and Webber, 1973):** a problem that is hard to define, has many interdependent causes, no single right answer and no clear stopping point, such as poverty, traffic or climate change.
- **Innovation:** turning a new idea into something of value that people adopt (product, service, process or business model).
- **Creativity:** the ability to generate ideas that are both new and useful.
- **Design mindset:** the attitudes of a design thinker: empathy, curiosity, optimism, comfort with ambiguity, bias towards action, learning from failure, collaboration.
- **Prototype:** a quick, cheap, rough model of an idea built to learn from users.
- **Iteration:** repeating cycles of building, testing and improving.
- **Desirability, feasibility, viability (IDEO):** a good solution is wanted by people, possible with technology, and sustainable for the business.

💡 Core concepts

Five stages of design thinking (Stanford d.school)

Stage	Purpose	Tools
Empathize	Understand users' experiences, needs and feelings	Interviews, observation, immersion, empathy maps
Define	Frame the right problem from insights	POV statements, personas, How Might We questions
Ideate	Generate many possible solutions	Brainstorming, mind maps, SCAMPER, Six Thinking Hats
Prototype	Make ideas tangible quickly	Paper models, storyboards, role play, wireframes
Test	Learn from users and refine	User testing, feedback grids, A/B tests

The stages are not strictly linear: teams loop back (for example from Test to Define) as they learn.

Traditional problem solving vs design thinking

Traditional (analytical)	Design thinking
Starts from the problem as given	Starts from the people affected and reframes the problem

Traditional (analytical)	Design thinking
Linear: analyse, plan, execute	Iterative: build, test, learn
Seeks the one right answer	Explores many possible answers
Avoids failure	Fails early and cheaply to learn
Expert-driven	Collaborative, multidisciplinary, involves users
Convergent thinking dominates	Alternates divergent and convergent thinking

History of design thinking: Herbert Simon's "The Sciences of the Artificial" (1969) described design as a way of thinking; Rittel and Webber defined wicked problems (1973); Bryan Lawson and Nigel Cross studied how designers think (1980s); Peter Rowe's book "Design Thinking" (1987); IDEO (David Kelley, Tim Brown) and the Stanford d.school (2005) made it popular in business and education; Roger Martin linked it to business strategy (2009).

Innovation and creativity in organizations and teams: creativity needs psychological safety, diverse teams, time and space to explore, tolerance of failure, and leaders who reward experiments. Barriers: fear of failure, rigid hierarchy, "we have always done it this way", functional silos, short-term pressure. Types of innovation: incremental, radical, disruptive; product, process, service, business model.

Elements of design: line, shape, form, space, colour, texture, value (light and dark), typography.

Principles of design: balance (symmetric, asymmetric), contrast, emphasis (focal point), movement, proportion and scale, hierarchy, repetition and rhythm, pattern, alignment, white space, unity.

Design mindset ("13 musical notes" in the AKTU course): the course uses a set of thirteen mindset traits, compared to musical notes, that together make a design thinker. A useful way to write them: empathy, curiosity, collaboration, optimism, experimentation, bias to action, embracing ambiguity, learning from failure, mindfulness of process, user-centredness, iteration, visual thinking and storytelling. Learn them as a list and give one example for each.

Examples of great design: the Jaipur Foot (low-cost prosthetic designed around users' squatting and barefoot walking), the Aravind Eye Hospital model, UPI and BHIM (simple payments for first-time users), OXO Good Grips peelers (designed for people with arthritis, used by everyone), Apple iPod click wheel, Airbnb's early redesign of listing photos, Embrace infant warmer (low-cost warmer for rural hospitals).

Design approaches across the world: human-centred design (IDEO, USA), Scandinavian participatory design (users co-design), Japanese kaizen and monozukuri (craft and continuous improvement), German Bauhaus and Dieter Rams' ten principles ("less, but better"), Indian jugaad (frugal innovation), British Design Council's double diamond.

Examples and comparisons

Example 1: Wicked vs tame problem. Fixing a broken projector is tame: clear problem, known solution, you know when it is done. Reducing student stress before exams is wicked: many causes (workload, family expectations, sleep, social media), stakeholders disagree, every intervention changes the situation, and there is no final solution.

Example 2: Five stages on a canteen queue. Empathize: observe the lunch rush and interview students; find that most waiting happens at the payment counter. Define: "Busy students need a faster way to pay because they lose half their break in the queue." Ideate: UPI QR at every counter, pre-order app, token

system, combo meals. Prototype: a printed QR code and a Google Form pre-order for one week. Test: measure queue time and get feedback; refine.

Key points to remember

- Five stages: Empathize, Define, Ideate, Prototype, Test (iterative).
- Wicked problems: no definitive formulation, no stopping rule, solutions are better or worse, not right or wrong.
- Desirability, feasibility, viability.
- Elements (line, shape, colour...) vs principles (balance, contrast, hierarchy...).

⭐ Important questions

1. What is design thinking? Compare traditional problem solving with design thinking.
2. Explain the history of design thinking and the concept of wicked problems.
3. Explain innovation and creativity in organizations and teams. What is the design mindset?
4. Explain the elements and principles of design with examples.
5. Give examples of great design and explain design approaches across the world.

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UNIT 2 Human Needs, Values and Empathy

✍️ 2-mark definitions

- **Self and body:** a human being is a coexistence of the self ("I", the conscious entity with desires, thoughts and feelings) and the body (the physical instrument). The self's needs (happiness) are qualitative and continuous; the body's needs (food, shelter) are quantitative and temporary.
- **Prosperity:** the feeling of having more than enough physical facilities, as opposed to wealth, which is only the facilities themselves.
- **Actualization:** realizing one's potential and aspirations (top of Maslow's hierarchy: self-actualization).
- **Socialization:** the process by which a person learns the values, norms and behaviours of their family, society and institutions.
- **Core values:** deep, lasting beliefs that guide behaviour (trust, respect, care, honesty).
- **Empathy:** understanding and sharing another person's feelings and perspective, "walking in their shoes".
- **Stakeholder:** any person or group affected by or able to influence a problem or its solution.
- **Empathy map:** a four-quadrant chart of what a user Says, Thinks, Does and Feels.
- **Customer journey map:** a visual timeline of every step a user goes through, with actions, touchpoints, emotions and pain points.
- **Insight:** a non-obvious understanding about users' needs or motivations, drawn from research, that can inspire design.

💡 Core concepts

From basic needs to actualization (Maslow's hierarchy): physiological → safety → love and belonging → esteem → self-actualization (later: self-transcendence). Design that serves only lower needs makes products; design that serves higher needs builds meaning.

Gap between desires and actualization: people often confuse the self's need for happiness with the body's need for things, so they chase more possessions without feeling prosperous. Understanding real needs (not just wants) is the basis of good design and of a meaningful life.

Culture in family, society, institutions and startups: culture is shared values and behaviour. In families it shapes trust and care; in institutions it shapes rules and roles; in startups it shapes speed, risk-taking and openness to failure. Design must fit the user's culture (for example a payment app for rural users must work in local languages and with low literacy).

Ethical human conduct: behaviour based on right understanding and right feelings in relationships (trust, respect, affection, care, guidance, reverence, glory, gratitude, love), leading to mutual happiness. Negative sentiments (anger, jealousy, fear, greed) block empathy.

Universal human goal: right understanding and happiness in the individual; prosperity in the family; fearlessness (trust) in society; co-existence with nature.

Identifying stakeholders: list users, customers, decision makers, influencers, suppliers, regulators, the community; map them on a power-interest grid (manage closely, keep satisfied, keep informed, monitor). Classify insights by stakeholder and by need.

Empathy tools

Tool	How it works
Interviews	Open-ended questions ("tell me about the last time..."), ask "why?", listen more than talk
Observation	Watch people in their real setting without interfering
Immersion	Experience the situation yourself (use a wheelchair on campus for a day)
Empathy map	Says, Thinks, Does, Feels (+ Pains and Gains)
Emotional mapping	Plot the user's emotional highs and lows across an experience
Customer journey map	Stages, actions, touchpoints, thoughts, emotions, pain points, opportunities
Brainstorming on insights	Group generates and clusters insights (affinity diagram)

Brainstorming do's and don'ts (IDEO rules): defer judgement, encourage wild ideas, build on others' ideas ("yes, and"), stay focused on the topic, one conversation at a time, be visual, go for quantity. Don'ts: criticize, let one person dominate, stop at the first good idea.

Moccasin walk activity: based on the saying "walk a mile in someone's moccasins": participants take on the role or experience of another person (for example a canteen worker, a visually impaired student) and then share what they felt, to build empathy.

Examples and comparisons

Example 1: Empathy map of a first-year hostel student.

Says	Thinks
"The Wi-Fi never works in my room." "Mess food is the same every day."	"Will I make friends?" "I hope I can cope with the syllabus."
Does	**Feels**
Studies in the corridor for signal, orders food online, calls home every night	Lonely, anxious, tired, sometimes excited

Pains: poor connectivity, homesickness, food. Gains: independence, new friends. Insight: the corridor has become an informal social space; a well-lit study lounge with good Wi-Fi could address both connectivity and loneliness.

Example 2: Customer journey of a patient at a government hospital OPD.

Stage	Action	Emotion	Pain point
Arrival	Finds the registration counter	Confused	No signs
Registration	Waits in queue	Frustrated	1-hour queue

Stage	Action	Emotion	Pain point
Waiting	Sits outside doctor's room	Anxious	No information on turn
Consultation	3-minute talk	Rushed	Cannot ask questions
Pharmacy	Another queue	Tired	Medicines out of stock

Opportunities: token display screens, online registration (ABHA), printed instructions, pharmacy stock alerts.

Key points to remember

- Self vs body needs; prosperity vs wealth; Maslow.
- Universal human goal: understanding, prosperity, trust, co-existence.
- Stakeholder mapping with the power-interest grid.
- Empathy map (Says, Thinks, Does, Feels); journey map; emotional map; immersion; Moccasin walk.
- Brainstorming rules: defer judgement, quantity, build on ideas.

⭐ Important questions

1. Explain the human being as the coexistence of self and body. Explain the gap between desires and actualization.
2. Explain the role of culture, values and ethical conduct in design.
3. How are stakeholders identified and classified?
4. Explain empathy tools: interviews, empathy maps, emotional mapping, immersion and observation, customer journey maps.
5. Explain brainstorming and its do's and don'ts.

UNIT 3 Problem Definition and Ideation

✍️ 2-mark definitions

- **Problem statement:** a clear, concise description of the user's problem to be solved, without suggesting a solution.
- **Persona:** a fictional but research-based character representing a key user group, with goals, behaviours, frustrations and context.
- **POV (Point of View) statement:** "[User] needs [need] because [insight]."
- **How Might We (HMW):** a question that turns a problem into an opportunity for ideas: "How might we ...?"
- **Double diamond:** the UK Design Council model with four phases: Discover (diverge), Define (converge), Develop (diverge), Deliver (converge).
- **Divergent thinking:** generating many different ideas.
- **Convergent thinking:** narrowing down to the best idea.
- **5 Whys:** asking "why?" repeatedly (about five times) to reach the root cause of a problem.
- **4 W's (and H):** analysing a problem by asking Who, What, Where, When (and Why, How).
- **Mind mapping:** a radial diagram with the central idea in the middle and branches for related ideas.
- **Six Thinking Hats (Edward de Bono):** a method of looking at a problem from six perspectives in turn, each represented by a coloured hat.
- **Random association:** linking the problem with a random word or object to spark new ideas.

💡 Core concepts

Good problem statement: human-centred, broad enough for creativity, narrow enough to be manageable, based on insights, free of a built-in solution.

Research for design: drivers (why we research), information gathering (primary: interviews, surveys, observation; secondary: reports, data), target groups and samples (who and how many), feedback loops. Mix qualitative (why) and quantitative (how many).

Idea generation process: set design directions and themes, collect inspirations (mood boards, analogies), brainstorm, sketch, present, evaluate (dot voting, impact-effort matrix, desirability-feasibility-viability).

Ideation tools

Tool	Use
Brainstorming / brainwriting	Many ideas fast; brainwriting (6-3-5) lets quiet members contribute
SCAMPER	Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse
Mind mapping	Explore connections around a central theme
Metaphor and random association	Break fixed thinking by linking to unrelated things

Tool	Use
Six Thinking Hats	Structured group thinking from six angles
Worst possible idea	Generate bad ideas, then reverse them
Million-Dollar Idea	Each person has a pretend budget to "invest" in ideas, to prioritize
Visual collaboration tools	Mural, Miro, Google Jamboard (now retired) for remote teams

Six Thinking Hats

Hat	Thinking
White	Facts and information
Red	Feelings, intuition, emotions
Black	Caution, risks, what can go wrong
Yellow	Optimism, benefits, value
Green	Creativity, new ideas, alternatives
Blue	Process control, summary, next steps (the chairperson's hat)

Ice-cream stick problem definition activity: teams are given ice-cream sticks and a vague brief; they must first define the actual problem (who it is for, what need) before building, showing how framing changes the solution.

Examples and comparisons

Example 1: Persona. "Riya, 19, B.Tech second year, day scholar from Ghaziabad. Travels 1.5 hours each way by bus and metro. Goal: keep up with lectures and placements prep. Frustration: notes are scattered across WhatsApp groups and PDFs she cannot read on her phone. Uses her phone for everything; data pack 1.5 GB/day."

Example 2: From insight to POV to HMW.

- Insight: students revise on the phone while commuting but most notes are scanned A4 PDFs that are hard to read on small screens.
- POV: "Commuting engineering students need exam notes that are easy to read on a phone because their revision time is on the bus and metro."
- HMW: "How might we make revision notes readable on a phone?" "How might we let students revise offline during the commute?" "How might we turn a one-hour commute into focused revision?"

Example 3: 5 Whys. Problem: students submit lab files late.

1. Why? They finish the experiments late.
2. Why? The lab computers are often not working.
3. Why? Software licences have expired.
4. Why? Nobody tracks licence renewal dates.
5. Why? There is no assigned owner for lab IT assets.

Root cause: no owner for lab IT; solution: assign responsibility and a renewal calendar.

Example 4: Six Thinking Hats on "Should the college allow mobile phones in class?" White: 90% of students carry smartphones; some teachers use online quizzes. Red: teachers feel ignored; students feel trusted. Black: distraction, cheating. Yellow: instant look-up, polls, recordings for revision. Green: "phone parking" with use only during set activities. Blue: summarize and agree on a trial policy for one semester.

Key points to remember

- POV = user + need + insight; HMW turns it into a question.
- Double diamond: Discover, Define, Develop, Deliver.
- Tools: 4W, 5 Whys, mind map, metaphor, random association, SCAMPER, Six Thinking Hats, Million-Dollar Idea.
- Six hats: white facts, red feelings, black caution, yellow benefits, green ideas, blue process.

⭐ Important questions

1. How do you define a problem statement? Explain personas and POV statements with an example.
2. Explain the role of research in design: drivers, information gathering, target groups, samples and feedback.
3. Explain the idea generation process and the double diamond approach.
4. Explain the 4 W's, 5 Whys and How Might We with examples.
5. Explain mind mapping, metaphor, random association and Six Thinking Hats.

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UNIT 4 Critical Thinking

✍️ 2-mark definitions

- **Critical thinking:** the disciplined process of actively analysing, evaluating and synthesizing information to reach well-reasoned judgements and decisions.
- **Ordinary thinking:** everyday, quick, often automatic thinking that accepts information without examining it.
- **Argument (in logic):** a set of statements (premises) offered to support a conclusion.
- **Rhetoric:** the art of persuasion, which may use emotion, style and authority rather than evidence.
- **Cognitive bias:** a systematic error in thinking that affects judgements (confirmation bias, anchoring, availability).
- **Tribalism:** favouring the views of one's own group and rejecting others' regardless of evidence.
- **Confirmation bias:** seeking and favouring information that supports what one already believes.
- **Incongruence:** a mismatch or inconsistency between ideas, statements or evidence.

💡 Core concepts

Critical vs ordinary thinking

Ordinary thinking	Critical thinking
Guesses	Estimates with evidence
Prefers	Evaluates
Groups by association	Classifies by criteria
Believes	Assumes and tests
Infers loosely	Infers logically
Accepts without question	Asks "how do we know?"
Judges without criteria	Judges with criteria

Traits of critical thinkers: curiosity, open-mindedness, scepticism (healthy doubt), intellectual humility, fairness, persistence, self-awareness of biases, clarity, willingness to change their mind with evidence.

Critical thinking skills: identifying arguments, linking ideas, structuring arguments, analysing and evaluating evidence, recognizing incongruences and inconsistencies, identifying assumptions, drawing conclusions, reflecting on one's own reasoning.

Five pillars of critical thinking (a common framework): (1) logic and reasoning, (2) evidence and sources, (3) awareness of assumptions and biases, (4) consideration of alternative perspectives, (5) clear conclusions and reflection. (Different books list slightly different pillars; the idea is to show that critical thinking rests on reasoning, evidence, self-awareness, open-mindedness and judgement.)

Argumentation vs rhetoric

Argumentation	Rhetoric
Aims at truth	Aims at persuasion
Uses evidence and logic (logos)	Also uses emotion (pathos) and authority (ethos)
Open to counter-arguments	May hide weaknesses
Judged by validity and soundness	Judged by effect on audience

Common cognitive biases

Bias	Example
Confirmation bias	Reading only reviews that praise the phone you want
Anchoring	First price quoted (₹2,000) makes ₹1,500 look cheap
Availability heuristic	Fearing flying after hearing of one crash
Bandwagon effect	Choosing a branch because friends did
Dunning-Kruger effect	Beginners overestimate their skill
Sunk cost fallacy	Continuing a failing project because money was already spent
Halo effect	Assuming a good speaker is also a good engineer

Politics and tribalism: group identity makes people accept claims from their side and reject the other's. Remedies: judge claims by evidence, steelman (state the strongest version of) the opposing view, check multiple sources.

Case study (applying critical thinking): a viral message claims "drinking hot water cures a virus". Critical thinker asks: who is the source? Is there evidence from controlled studies? Does it come from a credible health body (WHO, ICMR)? Is the claim specific and testable? Is there a motive to spread it? Result: no evidence, so do not forward.

Key points to remember

- Critical thinking = analyse + evaluate + decide with evidence.
- Traits: curiosity, open-mindedness, scepticism, humility.
- Skills: linking ideas, structuring arguments, spotting incongruence.
- Argumentation vs rhetoric (logos, pathos, ethos).
- Biases: confirmation, anchoring, availability, bandwagon, sunk cost, halo.

⭐ Important questions

1. What is critical thinking? Compare critical and ordinary thinking.
2. Explain the traits and skills of a critical thinker.
3. Explain the five pillars of critical thinking.
4. Distinguish argumentation and rhetoric.
5. Explain cognitive biases, tribalism and politics with examples. Discuss a case study.

UNIT 5 Arguments, Logic and Reasoning

✍️ 2-mark definitions

- **Statement (proposition):** a sentence that is either true or false.
- **Claim:** a statement someone asserts to be true.
- **Premise:** a statement given as a reason in an argument.
- **Conclusion:** the statement the premises are meant to support.
- **Valid argument:** an argument in which, if the premises are true, the conclusion must be true (form guarantees it).
- **Sound argument:** a valid argument whose premises are actually true.
- **Strong (inductive) argument:** one whose premises make the conclusion probable; cogent if also the premises are true.
- **Deductive reasoning:** reasoning from general premises to a certain conclusion.
- **Inductive reasoning:** reasoning from specific observations to a probable general conclusion.
- **Abductive reasoning:** inference to the best explanation.
- **Logical fallacy:** an error in reasoning that makes an argument invalid or weak.
- **Propositional logic:** logic of whole statements joined by connectives: NOT (¬), AND (∧), OR (∨), IF-THEN (→), IF AND ONLY IF (↔).

💡 Core concepts

Indicator words: premises: since, because, given that, as; conclusions: therefore, so, hence, thus, consequently.

Argument diagrams: number each statement; draw arrows from premises to conclusion; premises that work only together are joined (linked), premises that each support independently get separate arrows (convergent); a sub-conclusion can become a premise for the main conclusion (serial).

Valid deductive forms

Form	Structure
Modus ponens	If P then Q; P; therefore Q
Modus tollens	If P then Q; not Q; therefore not P
Hypothetical syllogism	If P then Q; if Q then R; therefore if P then R
Disjunctive syllogism	P or Q; not P; therefore Q
Categorical syllogism	All A are B; all B are C; therefore all A are C

Invalid forms: affirming the consequent (If P then Q; Q; therefore P) and denying the antecedent (If P then Q; not P; therefore not Q).

Logical vs scientific reasoning: logical reasoning checks whether conclusions follow from premises; scientific reasoning forms hypotheses, tests them by observation and experiment, and accepts

conclusions provisionally (falsifiability, controls, replication).

Common fallacies

Fallacy	Meaning	Example
Ad hominem	Attacking the person, not the argument	"Don't listen to his idea, he failed maths."
Straw man	Misrepresenting an argument to attack it	"You want less homework, so you want students to do nothing."
False dilemma	Only two options when more exist	"Either we ban phones or students fail."
Slippery slope	Assuming one step leads to extreme results	"If we allow one late submission, nobody will submit on time."
Appeal to authority	Relying on an irrelevant authority	"A cricketer says this bike is safest."
Bandwagon	Popular means true	"Everyone uses it, so it must be good."
Hasty generalization	Conclusion from too few cases	"Two CSE students were rude, so all CSE students are rude."
Post hoc	After it, so because of it	"I wore my lucky shirt and passed."
Circular reasoning	Conclusion assumed in premise	"He is trustworthy because he says so."
Red herring	Distracting with an irrelevant point	Changing the topic when asked about results

Probability and judgement: people misjudge probabilities: base rate neglect, gambler's fallacy (thinking a coin "owes" heads), conjunction fallacy (P(A and B) cannot exceed P(A)). Use base rates and simple calculations.

Obstacles to critical thinking: biases, emotions, peer pressure, lack of information, overconfidence, time pressure, egocentrism and sociocentrism, closed-mindedness.

Examples and comparisons

Example 1: Validity vs soundness.

- "All metals conduct electricity. Copper is a metal. So copper conducts electricity." Valid and sound.
- "All birds can fly. Penguins are birds. So penguins can fly." Valid (the form is correct) but **unsound** (the first premise is false).
- "If it rains, the road is wet. The road is wet. So it rained." **Invalid** (affirming the consequent; the road could be wet from a tanker).

Example 2: Truth table for P → Q.

P	Q	P → Q
T	T	T
T	F	F
F	T	T

P	Q	P → Q
F	F	T

An implication is false only when the premise is true and the conclusion false.

Example 3: Checking modus tollens with a truth table. ((P → Q) ∧ ¬Q) → ¬P is true in all four rows, so it is a tautology and the form is valid.

P	Q	P → Q	¬Q	(P → Q) ∧ ¬Q	¬P	Whole
T	T	T	F	F	F	T
T	F	F	T	F	F	T
F	T	T	F	F	T	T
F	F	T	T	T	T	T

Example 4: Base rate. 1 in 100 students cheats. A detector flags 90% of cheaters and wrongly flags 10% of honest students. If a student is flagged, the chance they cheated = (0.01 × 0.9) / (0.01 × 0.9 + 0.99 × 0.1) = 0.009 / 0.108 = **8.3%**, not 90%. Judgement must consider the base rate.

Example 5: Argument diagram. (1) Our campus app crashes on low-end phones. (2) 60% of our students use low-end phones. (3) So most students cannot use the app reliably. (4) Therefore we should build a lighter version. Statements 1 and 2 are linked premises for 3 (sub-conclusion); 3 supports 4 (serial).

Key points to remember

- Valid (form) vs sound (valid + true premises); strong and cogent for induction.
- Modus ponens and tollens are valid; affirming the consequent and denying the antecedent are not.
- Fallacies table; P → Q false only for T, F.
- Probability traps: base rate, gambler's fallacy, conjunction fallacy.

⭐ Important questions

1. Explain arguments, claims, statements, premises and conclusions. Explain validity, soundness and strength with examples.
2. Explain deductive and inductive arguments and argument diagrams.
3. Compare logical and scientific reasoning.
4. Explain common logical fallacies with examples.
5. Explain propositional logic with truth tables. Check the validity of an argument form.
6. Explain probability and judgement and the obstacles to critical thinking.

Last-hour checklist

- ☐ Design thinking definition; five stages table; traditional vs design thinking.

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☐ History (Simon, Rittel, IDEO, d.school); wicked problems; innovation and creativity; design mindset traits.

☐ Elements and principles of design; great design examples; world approaches.

☐ Self and body; prosperity; Maslow; culture; universal human goal.

☐ Stakeholder mapping; empathy tools; empathy map example; journey map; brainstorming rules; Moccasin walk.

☐ Problem statement; persona; POV; HMW; research; double diamond.

☐ 4W, 5 Whys, mind map, metaphor, random association, SCAMPER, Six Thinking Hats, Million-Dollar Idea.

☐ Critical vs ordinary thinking; traits; skills; five pillars; argument vs rhetoric; biases; tribalism.

☐ Premises and conclusions; valid vs sound; deductive vs inductive; argument diagrams.

☐ Fallacies table; truth tables; modus tollens check; base-rate example (8.3%); obstacles.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	101 Design Methods: A Structured Approach for Driving Innovation in Your Organization	Vijay Kumar	John Wiley and Sons (2013)	WorldCat
2	Foundations of Ethics and Management	B. P. Banerjee	Excel Books (2005)	WorldCat
3	Basics Design 08: Design Thinking	Gavin Ambrose, Paul Harris	AVA Publishing (2010)	WorldCat
4	The Design of Business: Why Design Thinking Is the Next Competitive Advantage	Roger L. Martin	Harvard Business Press (2009)	WorldCat

Syllabus source: AKTU Open Elective syllabus for B.Tech final year, 2025-26 (copy hosted by an affiliated college, PDF)